

Business Data Management Project - Analysis of Revenue Trend and Sales Volume of a Local Multipurpose Store

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Data Collected

Date	Drinking Water bottles	Soda	Bourbon	...
Category	Beverages	Beverages	Biscuits	...
08-01-2022	18	18	14	...
08-02-2022	21	22	15	...
...				

Table: The data supplied has 29 columns, one for Date and remaining for different product items whose categories were mentioned in the row beneath. Data Pre-processing and cleaning and EDA on this cleaned data was performed using Microsoft Excel.

Cleansed Dataset

Date	Day	Month	Category	SKU	Item-Name	Price	Sales	Revenue
08-01-2022	Mon	Aug-22	Beverages	BE01	Drinking Water Bottle	Rs.20	18	Rs.360
08-01-2022	Mon	Aug-22	Beverages	BE02	Drinking Soda	Rs.15	18	Rs.270
08-01-2022	Mon	Aug-22	Biscuits	BI01	Bourbon	Rs.25	14	Rs.350
...								

Table: The data supplied has 29 columns, one for Date and remaining for different product items whose categories were mentioned in the row beneath. Data Pre-processing and cleaning and EDA on this cleaned data was performed using Microsoft Excel.

SKU Dictionary

Item Name	SKU
Drinking Water Bottle	BE01
Drinking Soda	BE02
Soft Drink (Glass Bottle)	BE03
Soft Drink (1L plastic bottle)	BE04
Soft (500 ml plastic bottle)	BE05
Bourbon	BI01
Marie Biscuits	BI02
Biskfarm Biscuits	BI03
Haldiram Bhujia	SN01
Bapuji Cake	SN02
Chips	SN03
Pack of bread	SN04
Body Lotion	COSM01
Deodorants	COSM02

Table: SKU Dictionary

Item Name	SKU
Face Cream	COSM03
Black Ink Pen	WMP01
Blue Ink Pen	WMP02
Blank Long Notebook	WMNT01
Lined Long Notebook	WMNT02
Lined Short Notebook	WMNT03
Blank Short Notebook	WMNT04
Cover File	WMF01
Envelope File	WMF02
Large Envelope File	WMF03
Lead Pencils (Set of 3)	WMPE01
Graphite Pencils (Set of 6)	WMPE02
Color Pencils Set	WMPE03
Apsara Eraser Set	WME01

Table: SKU Dictionary

Sales Pareto and Revenue Pareto

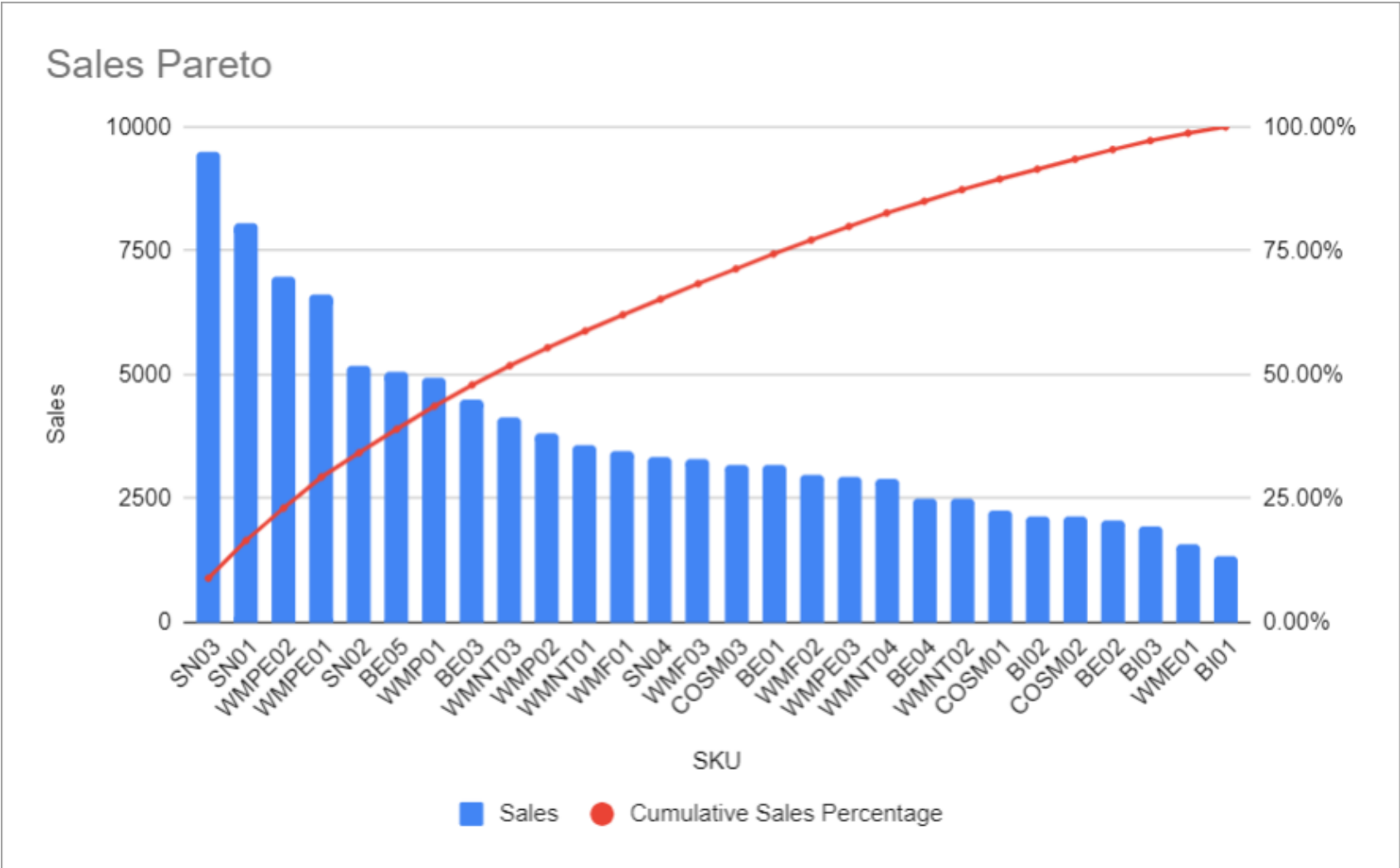


Figure: The Sales Pareto diagram

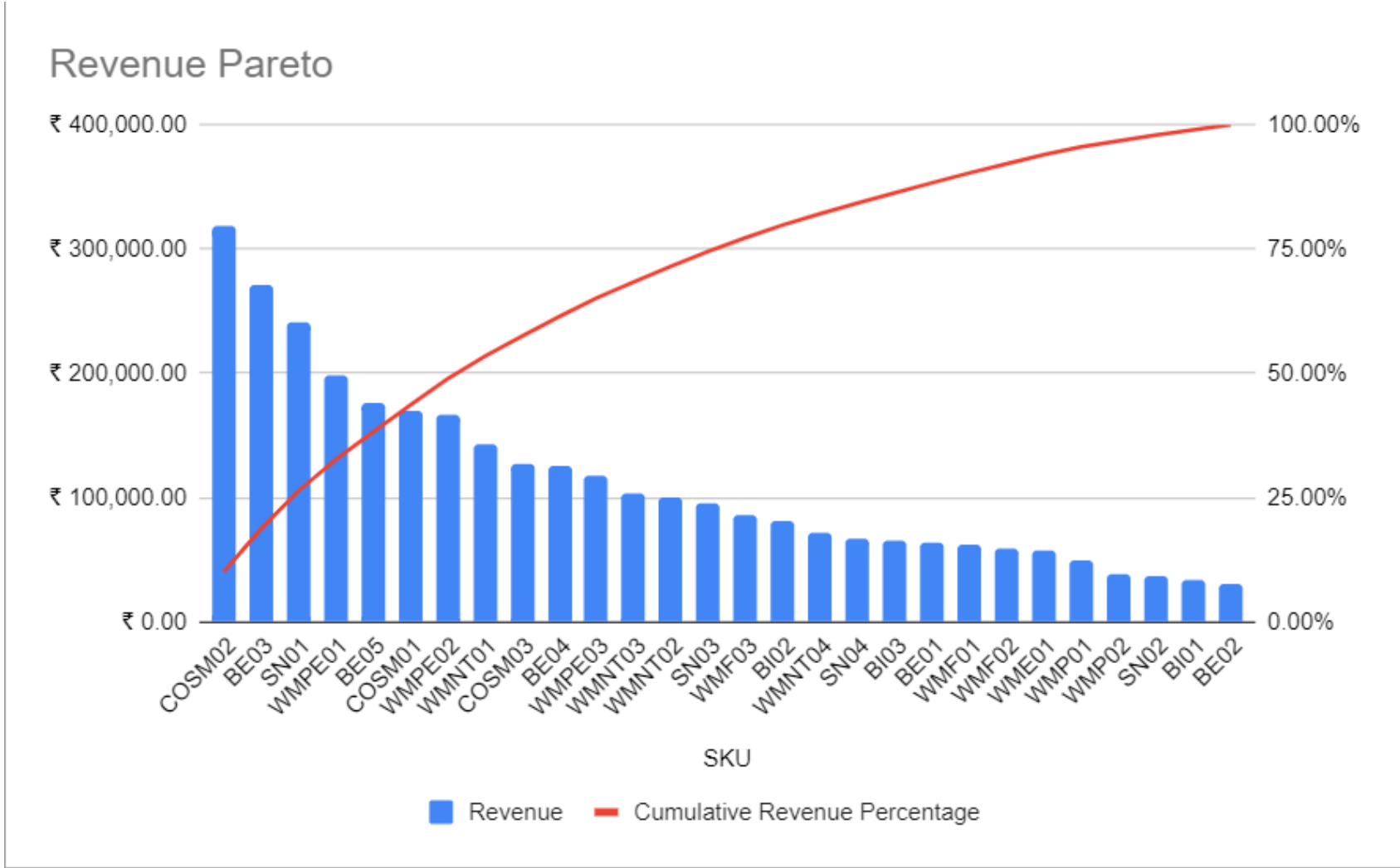
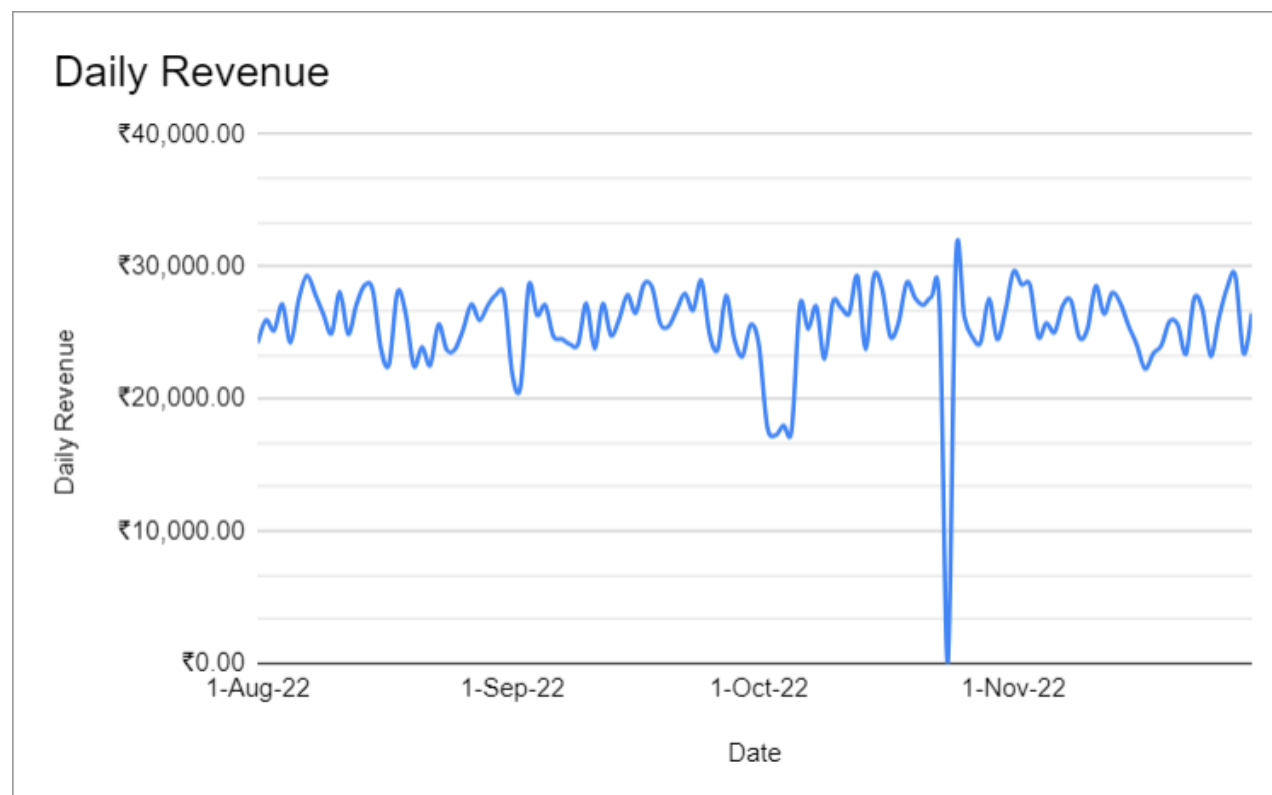


Figure: The revenue Pareto diagram

The pareto principle is not obeyed for both the figures

General Analysis on Sales Data/Revenue Data



Daily Sales and revenue trend

- The trend in daily sales and daily revenue proves to be significantly stable over the period of four months.
- Exceptions
 - We see a significant drop in the sales figure during the period 1st Oct 2022 to 5th Oct 2022. This is possibly due to the festival Durga-Puja, when the store remains open only for the first half of the day.
 - We see a reduction in the sales figure on 24th October 2022. Sales is zero on 24th October 2022 because the store was closed for the occasion of Kali-Puja.

Revenue generated per business unit

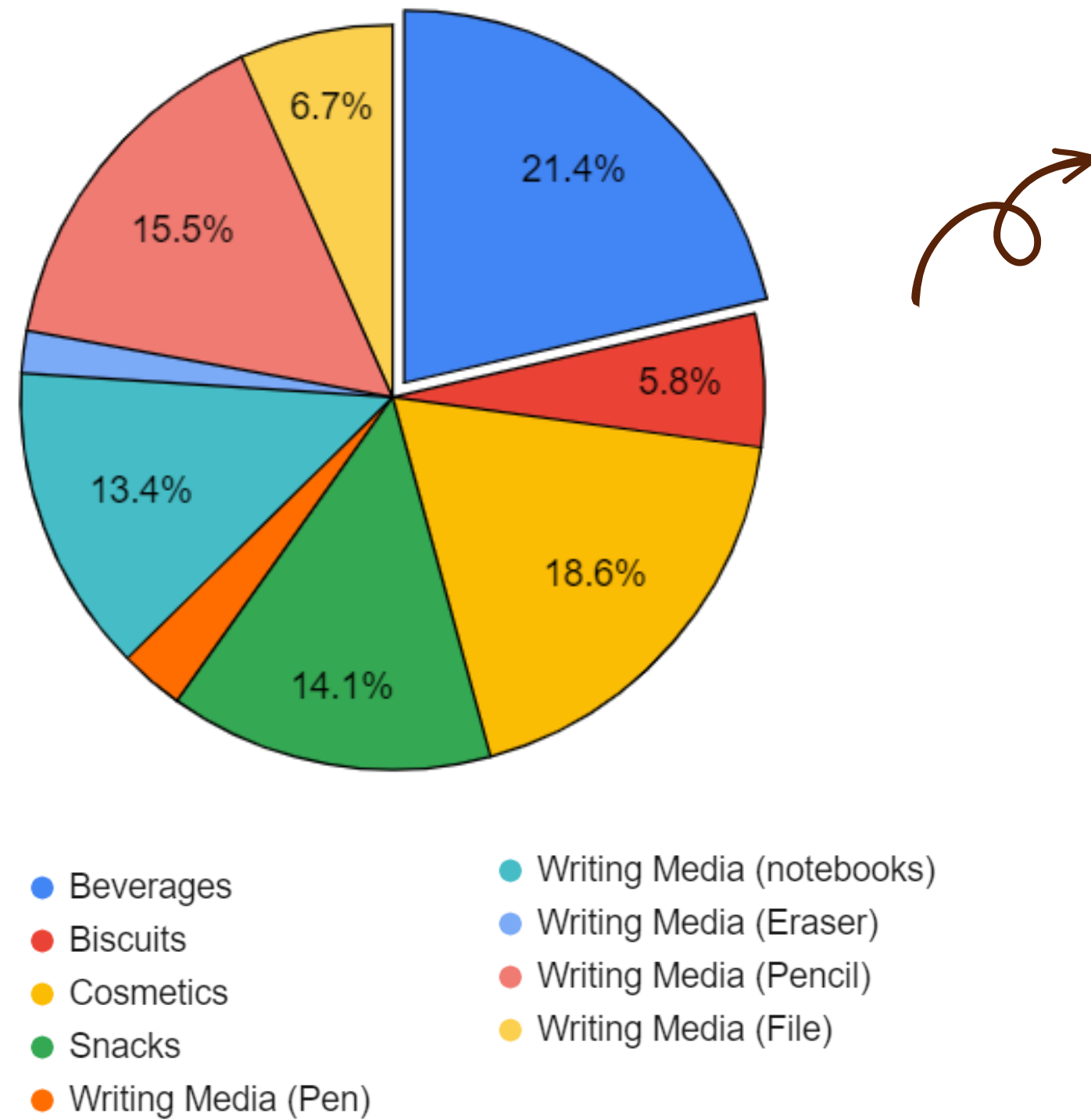


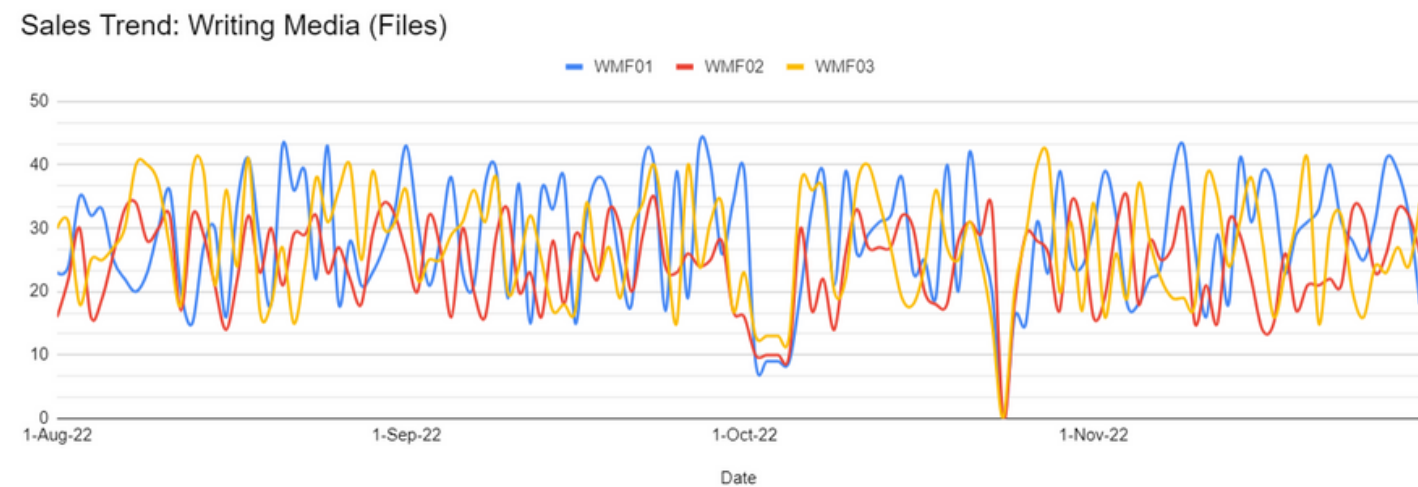
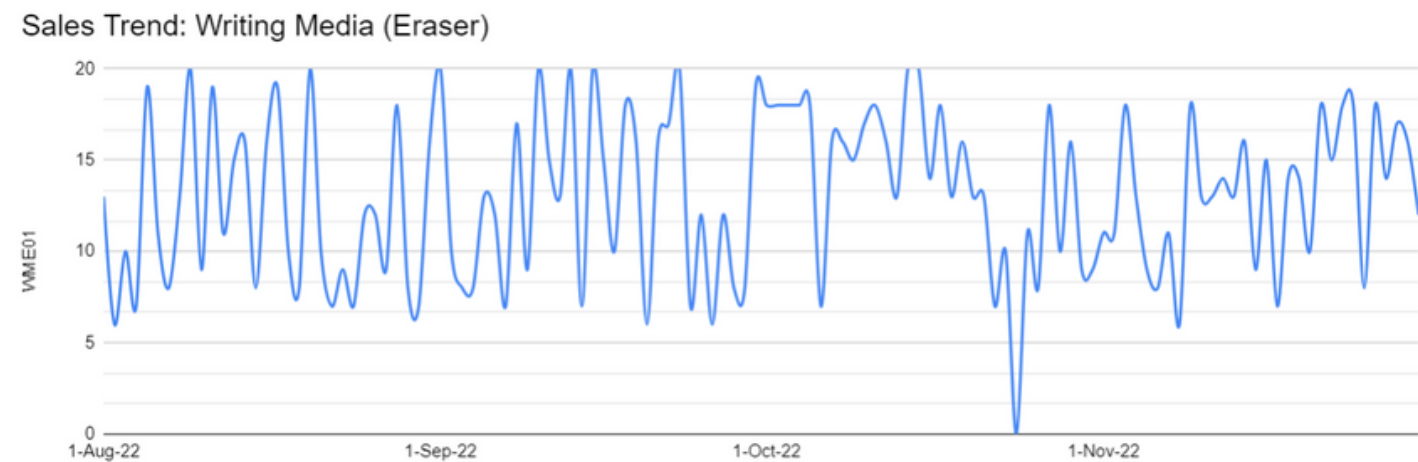
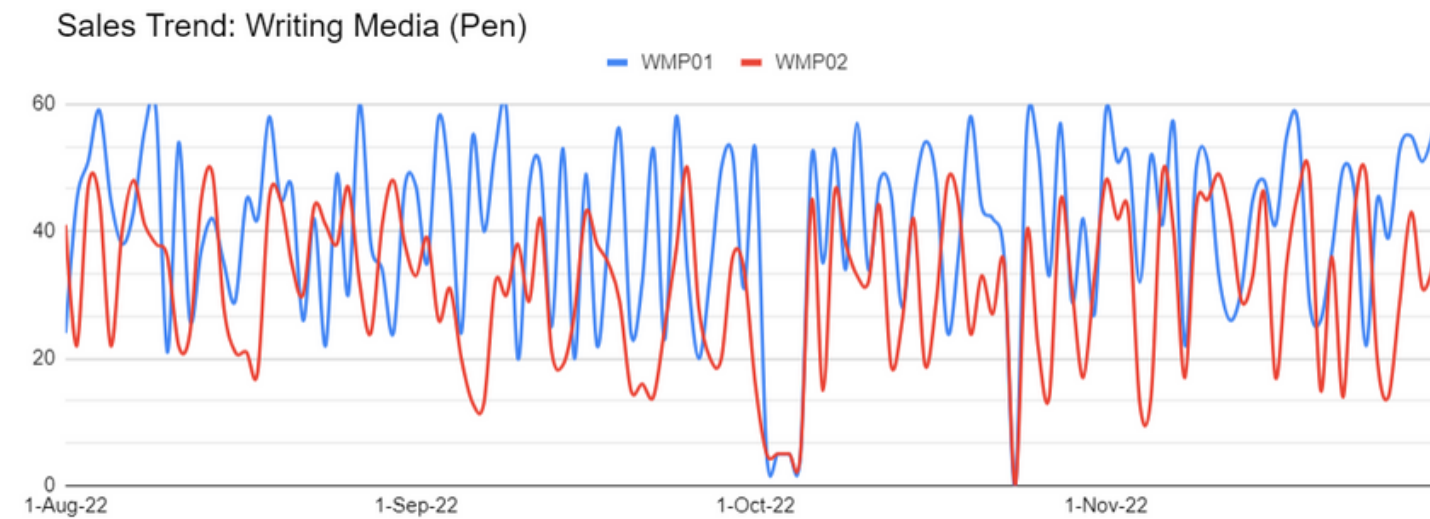
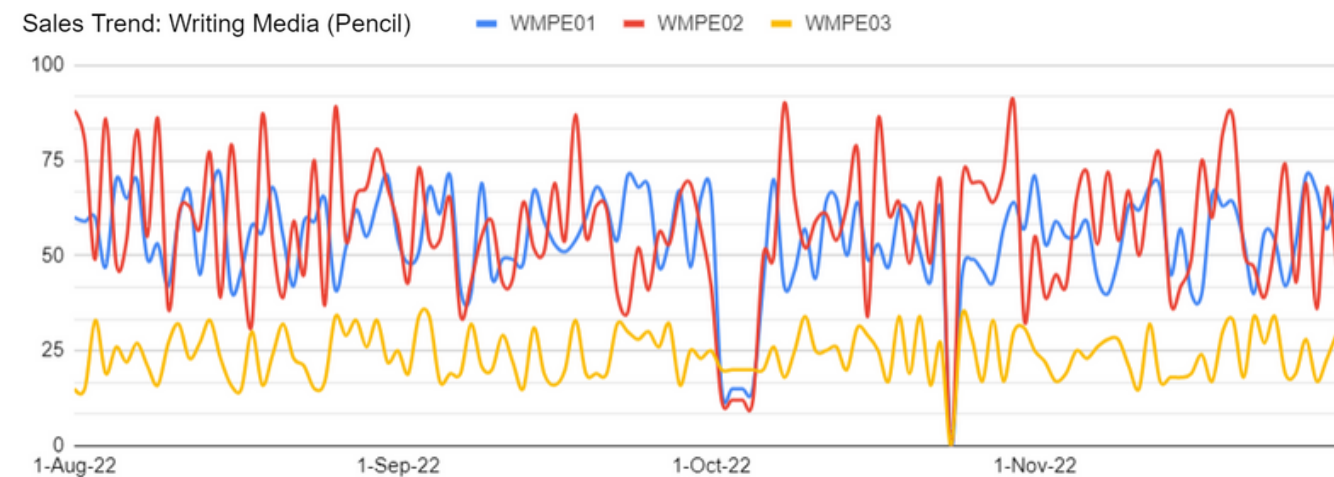
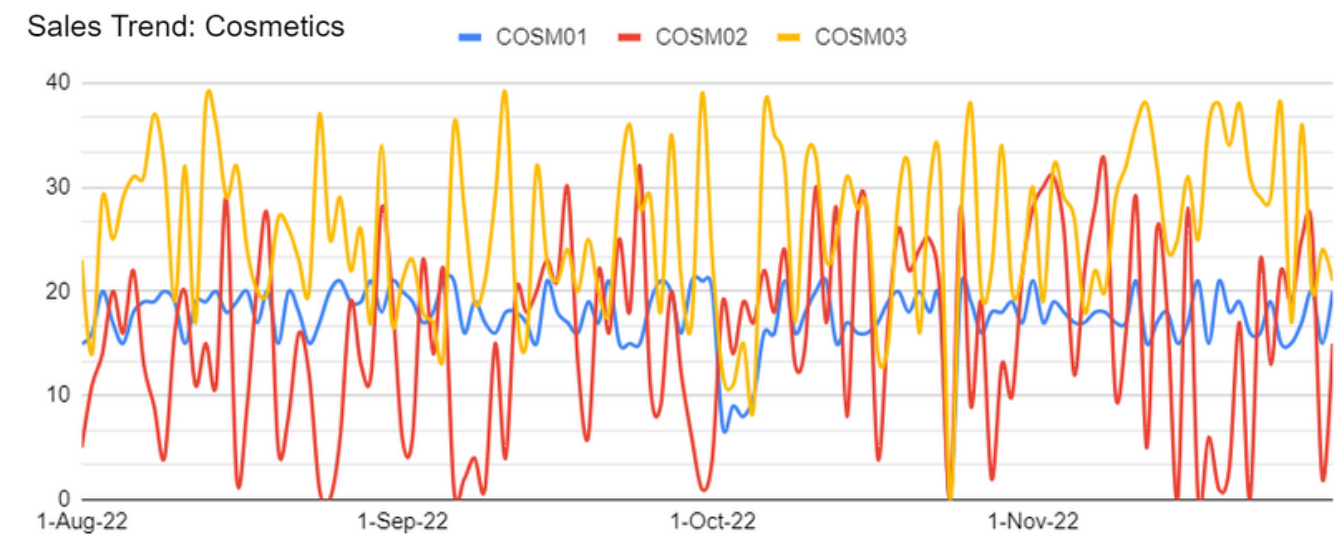
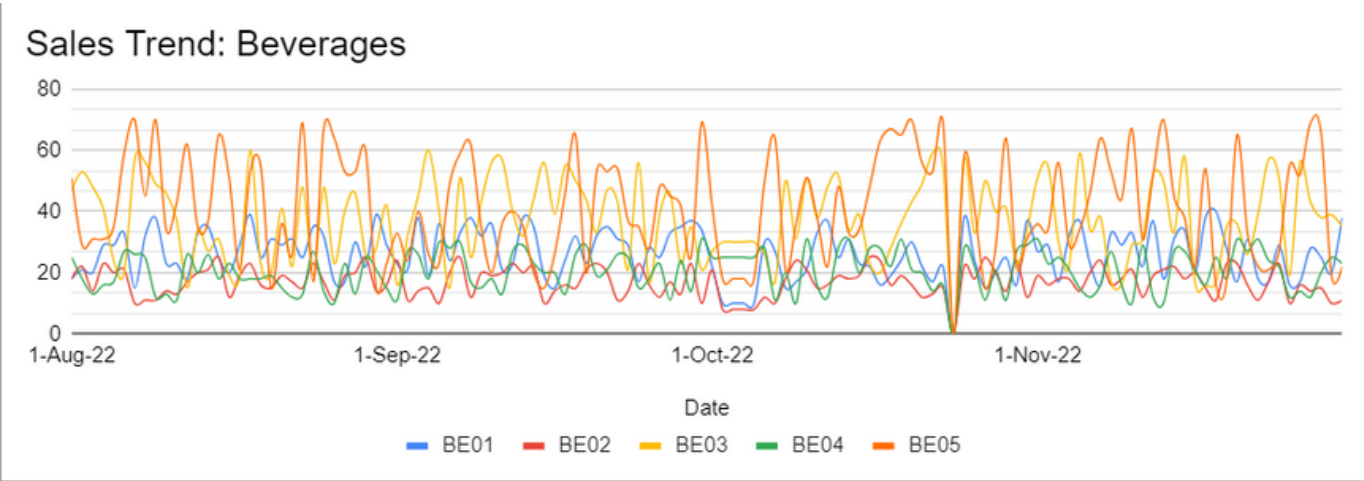
Figure: Proportion of revenue generated per business unit

This is the revenue contribution of each category at this store.

We find that Beverages is the highest in terms of revenue generation at this store followed by Cosmetic, Pencil, Snack, Notebook, File, Biscuits, Pen and Eraser.

Eraser contributes the least in the net revenue of this store.

Sales Trend: Sales trend pattern for important categories



Observations based on the trend patterns

- **BE05** is the most selling beverage at this store, followed by BE03, BE01, BE04, BE02 in that order. Since, BE02 remains quite low throughout, it is advisable to replace BE02 with a variant of BE05 or some other trending beverage in the market.
- **COSM03** is the most selling cosmetic product at this store followed by COSM01 and COSM02 in that order. Since we notice that the performance of COSM02 is very low throughout the period of four months, it is advisable to replace COSM02 with a variant of COSM03 or some other cosmetic product potentially capable of high sales in the current market.
- **WMPE02** is the most selling pencil at this store followed by WMPE01 and WMPE03. We notice that WMPE03 is significantly low during the entire period of four months. Hence, it is advisable to replace WMPE03 with a variant of WMPE02 or some new product potentially capable of high sales in the current market.
- **WMP01** is the better selling pen at this store than WMP02. We found that pen contribute less in the net revenue and there is no distinctive drop in the sales pattern, and hence introduction of variety of pens is advisable.
- There is only one product for **Erasers**. It has an oscillating trend pattern in its sales. Erasers contribute the least in the net revenue generation. We know pencils and erasers are complements; since pencils are one of the highest selling products at this store, it advisable to replace WME01 with a variety of erasers.
- **WMF01** is the highest selling file at this store. Since, there is no significant difference in sales with the other products WMF02 and WMF03. increasing the inventory space for this category maybe done to increase the net revenue generated.